The focus of this article is on Markov models for the analysis of panel data and, more specifically, on data obtained from repeated measurements of one categorical variable at several consecutive points in time. We first review developments in the field that attack the two main problems of the simple Markov model. The Mixed Markov model extends the simple model by allowing for population heterogeneity; the Latent Markov model incorporates measurement error and latent change into the simple model. Second, we present the more general Latent Mixed Markov model and show how both the Mixed Markov model and the Latent Markov model, as well as several more specific models, relate to this more general model. Finally, we reanalyze the Los Angeles panel data on depression with a focus on stability and change.

A Unifying Framework for Markov Modeling in Discrete Space and Discrete Time

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For many years, there has been considerable interest in modeling the stochastic processes underlying a set of categorical panel data. In what follows, we will (1) take a closer look at some developments in the field of Markov modeling, (2) present the more general Latent Mixed Markov model, and (3) report results from some reanalyses of the Los Angeles panel data on depression examined by Morgan et al. (1983).

We would like to stress that all of the models considered in this paper are discrete-time models. In addition, we consider first-order

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models with time homogeneous transition probabilities only, although incorporation of higher-order effects and time heterogeneity of transition probabilities are feasible in principle.

THE SIMPLE MARKOV CHAIN MODEL

In the simple Markov model, it is assumed that the process under consideration starts with a certain probability of occurrence of each of the R response categories (1, . . . ,i, . . . ,R) of the variable at time point t = 1. Given membership in category i at time t, a person may stay with this category i at time point t + 1 or switch to any of the remaining R – 1 categories. In this model, it is thus assumed that all persons in category i at time t have the same probabilities to be again in i at t + 1 or to switch, regardless of their history or any other attribute.

The model is therefore characterized by R – 1 initial probabilities and (T–1) (R(R–1)) conditional or transition probabilities (where T denotes the total number of time points under investigation). Because the T – 1 sets of transitions probabilities are assumed to be equal across time, or time homogeneous, the model has (R–1) + (R(R–1)) parameters.

Although this model seems conceptually interesting and efficient methods of estimation have been available since the Anderson-Goodman developments (Anderson, 1954; Anderson and Goodman, 1957; Goodman, 1962; see also Bishop et al., 1975; Langeheine, 1988a; Van de Pol and De Leeuw, 1986), the model suffers from the fact that it almost never fits the data (see, for example, Bartholomew, 1981; Coleman, 1964a; Langeheine, 1988a; Logan, 1981; Plewis, 1981; Wiggins, 1973). Several reasons have therefore been proposed in the literature that may be responsible for this misfit.

The two most prominent of these reasons are: (1) The model assumes population homogeneity though heterogeneity may be present (see, for example, Anderson, 1954; Blumen et al., 1955; Coleman, 1964a; Goodman, 1962; Wiggins, 1973); (2) the model does not allow for response or measurement error (see Bye and Schechter, 1986; Coleman, 1964a; Van de Pol and De Leeuw, 1986; Wiggins, 1973). We will comment on the first of these issues in the next two sections and on the second in the section after those.

INCORPORATING POPULATION HETEROGENEITY INTO A MARKOV CHAIN MODEL: THE MOVER-STAYER MODEL

Several authors have proposed to attack the heterogeneity problem by stratifying the total sample according to one or more discrete exogenous variables. The simple Markov model is fitted to each of these strata, and the results are compared either informally or by a formal test on (the equality of) the transition matrices. Although this approach is desirable on theoretical grounds, it is an indirect attack on the heterogeneity problem. To our knowledge, the first direct attack on the problem is due to Blumen et al. (1955). These authors proposed the Mover-Stayer model, which incorporates heterogeneity explicitly by assuming a mixture of two Markov chains: Movers are characterized by an ordinary Markov chain, whereas stayers stay in their initial category with probability 1 across time (that is, their transition matrix is equal to the identity matrix). Until recently, this model has been plagued by problems of parameter estimation. Goodman (1961) showed that the proportion of stayers is overestimated by the Blumen et al. procedure and presented consistent estimates. However, Morgan et al. (1983), as well as Frydman (1984), found that the Goodman estimates may differ from maximum likelihood estimates (MLEs) considerably. Morgan et al. used a derivative-free nonlinear regression procedure in estimating the parameters. Whereas this procedure supplies MLEs of the parameters of the Mover-Stayer model as well as of some constrained versions of this model, it can be rather laborious to implement the computations and, more important, the approach does not extend to more general types of models. Both Poulsen (1982) and Van de Pol and Langeheine (1989) therefore conceive of the Mover-Stayer model as a constrained version of the more general Mixed Markov model for which efficient MLEs are available.

INCORPORATING POPULATION HETEROGENEITY INTO A MARKOV CHAIN MODEL: POULSEN'S MIXED MARKOV CHAIN MODEL

Several authors (e.g., Anderson, 1954; Blumen et al., 1966; Spilerman, 1972) have imagined a general Mixed Markov model (i.e., a model that is a mixture of several ordinary Markov chains). To our knowledge, however, Poulsen (1982) was the first to introduce an exact formulation of the model, followed by Davies and Crouchley (1986), who restricted initial probabilities to be steady-state probabilities. Poulsen gives several references in which attempts were made to introduce population heterogeneity into a Markov model. As he notes, however, all of these attempts are limited in one way or the other. He therefore presents the general Mixed Markov (MM) model, which offers a direct attack on the problem of population heterogeneity.

Given measurements from a sample of N individuals at T = 3 points in time (or waves), the expected frequencies m_{ijk} (where i, j, and k refer to the R response categories at t = 1, t = 2, and t = 3) of a Mixed Markov model postulating S latent chains or classes (1, . . . , s, . . . , S) are given by

$$m_{ijk} = N \sum_{s=1}^{S} \pi_s \, \delta_{i\,|\,s} \, \tau_{j\,|\,is} \, \tau_{k\,|\,js} \qquad [1]$$

The model thus contains three kinds of parameters:

1. The π's are the proportions of each of the S chains.
2. The δ's are the initial probabilities at t = 1, given membership in a certain chain s.
3. The τ's are the transition probabilities at time t + 1 given time t and membership in chain s. In case of an R-categorical item, we thus have (S(T−1)) R × R matrices of transition probabilities, where the rows refer to the manifest category being conditioned on.

Because these are conditional probabilities, rows have to sum to unity. For the sake of simplicity, model definition is presented for T = 3 points in time only throughout. Extensions to more points in time are straightforward, however.

It should be noted that model (1) contains at least the following special cases as submodels:

1. If S = 1, we obtain the simple Markov chain model.
2. If S = 2 and one of the transition matrices is required to be the identity matrix, the result is the Blumen et al. (1955) Mover-Stayer model. Special versions of this model considered by Morgan et al. (1983) are obtained by imposing additional equality constraints on the transition

matrix of the movers. If all transition probabilities as well as initial probabilities of the movers are fixed according to equiprobability, we have the Converse (1964, 1970) Black & White model.

3. The MM model contains Lazarsfeld's (1950) classical Latent Class (LC) model as a special case if the rows of the transition matrices are constrained to be equal within classes. This result becomes evident immediately if the conditional indices i and j are omitted from the τ's in (1). This is equivalent to assuming local independence (within classes) between items (time points here; see Langeheine, 1988b). Local dependence is allowed for in (1) by means of the transition matrices; that is, by conditioning on the previous measurement.

Parameter estimation by use of the EM algorithm (Dempster et al., 1977) and problems of identifiability of the MM parameters are discussed in Poulsen (1982) and Van de Pol and Langeheine (1989). Evaluation of model fit may be done by either the likelihood ratio (LR) or Pearson (χ^2) chi-squared statistics.

In passing, we would like to stress some of the pros and cons of the MM model. First, it may be noted that the MM model is a direct extension of the simple Markov model into the latent class framework. Thus, the MM model offers a direct solution of the problem of population heterogeneity. Because MM models may be weakly identified or the EM algorithm may find a local maximum, we strongly advocate the use of several different sets of starting values together with a rather tight convergence criterion (see Van de Pol and Langeheine, 1989).

Second, the MM model is a relatively general model that contains a variety of other models well known in the literature as special cases; for example, Mover-Stayer, Black & White, and classical LC models.

Finally, it should be stressed that the model suffers from one serious drawback: Because class membership is assumed to be constant across time, the hypothesis of "real" change (i.e., switching from one class to another through time) may not be evaluated. This assumption will be relaxed by the model considered next.

INCORPORATION OF MEASUREMENT ERROR INTO A MARKOV CHAIN MODEL: THE LATENT MARKOV CHAIN MODEL

The Latent Markov (LM) model considered in this section is due to Wiggins (1955, 1973). Given a three-wave panel, the expected counts m_{ijk} of the Latent Markov model postulating C classes are obtained by

$$m_{ijk} = N \sum_{a=1}^{C} \sum_{b=1}^{C} \sum_{c=1}^{C} \delta_a \, \rho_{i|a} \, \tau_{b|a} \, \rho_{j|b} \, \tau_{c|b} \, \rho_{k|c} \qquad [2]$$

This model also contains three kinds of parameters:

1. The δ's are latent class proportions at $t = 1$.
2. The ρ's are conditional response probabilities at the three points in time, given class membership at the respective time points. Note that responses at each point in time are tied to its own latent variable.
3. The τ's are elements of the latent transition matrices at time $t + 1$ given time t.

To keep the model simple in terms of parameters, time homogeneity is assumed with respect to both the ρ's and the τ's (where at most $T - 2$ different sets of response probabilities are a necessary condition for identifiability; see Henry, 1973).

Unfortunately, the Wiggins approach suffers from several drawbacks (see Langeheine, 1988a, b; Poulsen, 1982; Van de Pol and De Leeuw, 1986). The most serious of these problems—inefficient parameter estimation—was surmounted by Poulsen and later Van de Pol and De Leeuw, who showed how to estimate the parameters of the LM model by use of the EM algorithm of Dempster et al. (1977). Bye and Schechter (1986) used another algorithm and forced the parameters within the 0 to 1 range by a transformation.

The main feature of this model may be seen in the fact that individuals are allowed to change class membership through time. It should be noted, however, that while population heterogeneity enters into the model by means of the ρ's, latent transitions are assumed to hold for all individuals. Thus, this model is a one-chain model because there is only one chain of latent transitions. Another feature of this model is that it allows for an error of measurements interpretation when the number of latent classes, C, equals the number of observed categories, R. In this case, the modal conditional response probabili-

TABLE 1: Restrictions on Latent Mixed Markov Chain Model (3) Leading to Specific Cases

	S	$\delta_{a\mid s}$	$\rho_{i\mid as}$	$\tau_{b\mid as}$
Latent class model	1	free	free	$a = b$: $\tau_{b\mid as} = 1$ $a \neq b$: $\tau_{b\mid as} = 0$
Latent Markov model	1	free	free	free
Single Markov chain	1	free	$a = i$: $\rho_{i\mid as} = 1$ $a \neq i$: $\rho_{i\mid as} = 0$	free
Mixed Markov model	≥2	free	$a = i$: $\rho_{i\mid as} = 1$ $a \neq i$: $\rho_{i\mid as} = 0$	free
Mover-Stayer model	2	free	$a = i$: $\rho_{i\mid as} = 1$ $a \neq i$: $\rho_{i\mid as} = 0$	$a = b$: $\tau_{b\mid a2} = 1$ $a \neq b$: $\tau_{b\mid a2} = 0$
Black & White model	2	$\delta_{a\mid 1} = 1/R$	$a = i$: $\rho_{i\mid as} = 1$ $a \neq i$: $\rho_{i\mid as} = 0$	$a = b$: $\tau_{b\mid a1} = 1/R$ $a \neq b$: $\tau_{b\mid a1} = 1/R$ $a = b$: $\tau_{b\mid a2} = 1$ $a \neq b$: $\tau_{b\mid a2} = 0$

NOTES: S = number of Markov chains.
$\delta_{a\mid s}$ = (latent) initial (wave 1) proportion of class a given chain s.
$\rho_{i\mid as}$ = probability of response i, given class a in chain s, for any time t.
$\tau_{b\mid as}$ = probability of being in class b at time t + 1, given class a at time t and chain s.

ties may be conceived of as reliabilities, whereas the nonmodal ones are considered as error rates.

As will be demonstrated in the Los Angeles data section, it is therefore possible to assess such quantities as true stability, true change, and error. Finally, we would like to mention that two simpler models are related to the LM model. The classical LC model is a constrained version of the LM model with the transition matrix fixed as an identity matrix. The fit obtained for the latter (no latent change) model should be compared with the fit obtained for model (2) before arguing in favor of latent change captured by the LM model. The simple Markov model is obtained if no measurement error is assumed.

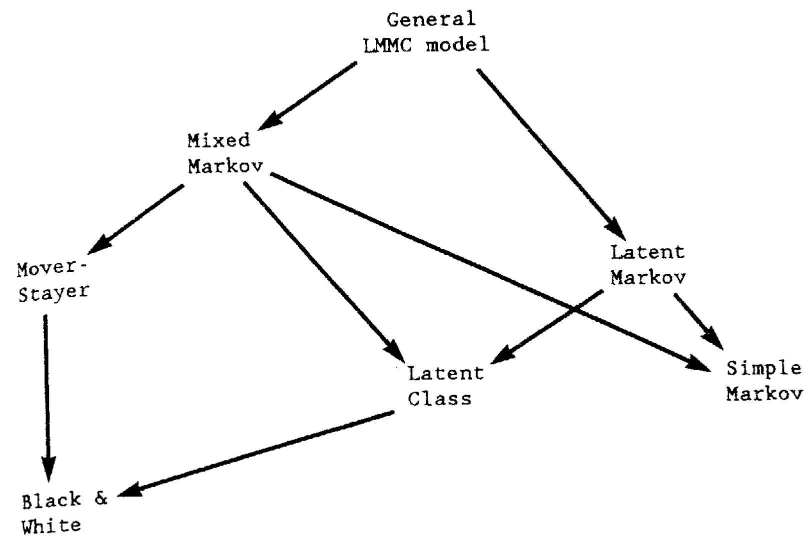

Figure 1: Hierarchical Relationships Between General LMMC Model (3) and Specific Cases (Table 1)

A LATENT MIXED MARKOV CHAIN MODEL

Whereas the aim of the previous sections was to give an overview of some developments in the field, the aim of this section is to bring together several Markov-type models into a single context, or to provide a unifying framework for Markov models.

The general Latent Mixed Markov Chain (LMMC) model is given by

$$m_{ijk} = N \sum_{s=1}^{S} \sum_{a=1}^{C} \sum_{b=1}^{C} \sum_{c=1}^{C} \pi_s\, \delta_{a\mid s}\, \rho_{i\mid as}\, \tau_{b\mid as}\, \rho_{j\mid bs}\, \tau_{c\mid bs}\, \rho_{k\mid cs} \qquad [3]$$

Comparison with model (2) reveals that we simply extend the one-chain LM model to an S-chain LM model; that is, to a mixture of S latent Markov chains. We introduce one additional set of parameters, π's, which are the proportions of the S latent chains. Within each chain, the chain variable, with index s, is added as a conditioning variable to

TABLE 2: Latent Transition Parameters* in 2-Chain, 2-Class Latent Markov Model (C′ = 2 × 2 Classes)

					Time t + 1				
				C′ (t + 1)	*1*	*2*	*3*	*4*	
				Chain s	*1*	*1*	*2*	*2*	
	Chain Class			*Class b*	*1*	*2*	*1*	*2*	*total*
	c′ (t)	*s*	*a*						
Time t	1	1	1		$\tau_{1\|11}$	$1-\tau_{1\|11}$	0	0	1
	2	1	2		$\tau_{1\|21}$	$1-\tau_{1\|21}$	0	0	1
	3	2	1		0	0	$\tau_{1\|12}$	$1-\tau_{1\|12}$	1
	4	2	2		0	0	$\tau_{1\|22}$	$1-\tau_{1\|22}$	1

*Zero parameters are fixed by definition.

the rest of the parameters. Table 1 shows which constraints have to be imposed on the general LMMC model to obtain simpler models. Figure 1 shows the relationships between specific LMMC models.

Because the extension from S = 1 to S ≥ 2 latent Markov chains is straightforward with respect to estimation of the parameters of the LMMC model by use of the EM algorithm of Dempster et al. (1977), we will not go into details here. As Table 2 shows, any S-chain C-class model may be set up as model with C′ = S × C classes with restrictions on certain latent transitions.

SOME REANALYSES OF THE LOS ANGELES PANEL DATA

For details on the Los Angeles panel data, the reader is referred to the Morgan et al. (1983) article. Suffice it to say that these data (see Table 3) refer to the depression state (0 = not depressed, 1 = depressed) available from four consecutive points in time for a sample of 752 respondents. In what follows, we first refer to the definition and fit of several models, and then discuss stability and change according to these models.

DEFINITION AND FIT OF SEVERAL MODELS

The seven models listed in Table 3 were all fitted to the data. Table 3 gives expected counts, likelihood ratio test statistics (LR), and degrees

TABLE 3: Observed and Expected Counts, Likelihood Ratio Test Statistics (LR), and Degrees of Freedom (df) for Several Models

*Time**				*Observed*	*Expected Counts for Model***						
t = 1	*t = 2*	*t = 3*	*t = 4*	*Counts*	*A*	*B*	*C*	*D*	*E*	*F*	*G*
0	0	0	0	487	487.00	487.00	487.00	485.80	485.63	486.00	487.02
0	0	0	1	35	33.32	31.71	35.04	35.25	38.56	33.27	31.52
0	0	1	0	27	30.58	31.71	35.04	33.59	38.56	34.39	33.40
0	0	1	1	6	13.03	11.55	13.94	9.01	9.28	7.23	8.22
0	1	0	0	39	30.58	31.71	35.04	33.59	38.56	37.66	37.72
0	1	0	1	11	9.44	11.55	13.94	6.96	9.28	7.44	8.45
0	1	1	0	9	11.96	11.55	13.94	8.53	9.28	9.34	10.49
0	1	1	1	7	5.09	4.21	5.55	8.27	10.35	9.19	7.54
1	0	0	0	50	47.12	44.96	35.04	51.53	38.56	47.15	47.60
1	0	0	1	11	14.54	16.38	13.94	10.03	9.28	8.03	8.97
1	0	1	0	9	13.34	16.38	13.94	9.50	9.28	9.93	11.01
1	0	1	1	9	5.69	5.97	5.55	10.34	10.35	9.23	7.57
1	1	0	0	16	18.42	16.38	13.94	12.45	9.28	15.45	15.66
1	1	0	1	9	5.69	5.97	5.55	10.34	10.35	9.57	7.81
1	1	1	0	8	7.20	5.97	5.55	10.99	10.35	12.78	10.01
1	1	1	1	19	19.00	19.00	19.00	15.82	15.05	15.34	19.00
Likelihood ratio:					15.50	17.87	24.39	8.54	16.40	8.32	5.16
Degrees of freedom:					10	11	12	8	12	10	8

*Categories refer to "not depressed" = 0 and "depressed" = 1.
**These models are A, Mover-Stayer model; B, Mover-Stayer model plus equality of rows of the mover transition probabilities; C, Mover-Stayer model with equilibrium assumption for movers; D, 2-chain Mixed Markov model; E, 2-class Latent Class model; F, 2-class Latent Markov model; and G, partially Latent Mover-Stayer model.

of freedom (df) for each of these models. The respective estimated parameter values may be found in Tables 4, 5, and 6. Results reported here have been obtained with PANMARK (Van de Pol et al., 1989), a PC program that uses the EM algorithm to obtain MLE's.[1] Models B, C and E can also be estimated with other programs for latent class analysis, such as MLLSA (Clogg, 1977), LAT (Haberman, 1979) or LCAG (Hagenaars and Luijkx, 1987).

Because Morgan et al. (1983) focus on the Mover-Stayer model, we begin with this model. As we have mentioned before, this model is a constrained version of a two-chain Mixed Markov model, where the movers follow a Markov process and the stayers stay in their initial category across time with probability 1. The likelihood ratio test statistic of 15.50 is identical to the one reported by Morgan et al. Because this model has 10 df, the fit may be considered adequate.

TABLE 4: Estimated Parameter Values* (with Standard Errors) for Models A through E Table 3

Model	Chain, Category	Chain Proportions, π's	Initial Probabilities, δ's	Transition Probabilities, τ's: t to t + 1, 0	t to t + 1, 1	t = 1 to t = 4, 0	t = 1 to t = 4, 1
A	chain 1,	.475, (.038)					
	0 = nondepressed		.678, (.034)	.764, (.027)	.236, (.027)	.748	.252
	1 = depressed		.322, (.034)	.701, (.030)	.299, (.030)	.748	.252
	chain 2,	.525, (.038)					
	0		.960, (.011)	1	0	1	0
	1		.040, (.011)	0	1	0	1
B	chain 1,	.446, (.028)					
	0		.659, (.032)	.733, (.020)	.267, (.020)	.733	.267
	1		.341, (.032)	.733, (.020)	.267, (.020)	.733	.267
	chain 2,	.554, (.028)					
	0		.960, (.011)	1	0	1	0
	1		.040, (.011)	0	1	0	1
C	chain 1,	.447, (.028)					
	0		.715, (.020)	.715, (.020)	.285, (.020)	.715	.285
	1		.285, (.020)	.715, (.020)	.285, (.020)	.715	.285
	chain 2,	.553, (.028)					
	0		.959, (.011)	1	0	1	0
	1		.041, (.011)	0	1	0	1
D	chain 1,	.854, (0.47)					
	0		.908, (.023)	.939, (.010)	.061, (.010)	.936	.064
	1		.092, (.023)	.895, (.075)	.105, (.075)	.936	.064
	chain 2,	.146, (.047)					
	0		.342, (.090)	.420, (.167)	.580, (.167)	.407	.593
	1		.658, (.090)	.398, (.060)	.602, (.060)	.407	.593
E	chain 1,	.842, (.025)					
	0		.934, (.008)	.934, (.008)	.066, (.008)	.934	.066
	1		.066, (.008)	.934, (.008)	.066, (.008)	.934	.066
	chain 2,	.158, (.025)					
	0		.404, (.044)	.404, (.044)	.596, (.044)	.404	.596
	1		.596, (.044)	.404, (.044)	.596, (.044)	.404	.596

*Parameter values of 0 and 1 are fixed by definition.

According to this model, 52.5% of the sample are classified to be stayers, 96% of which are perfectly stable nondepressed and 4% perfectly stable depressed respondents. Roughly two-thirds of the

TABLE 5: Estimated Parameter Values* (and Standard Errors) for Latent Markov Model, F

Class	Class Proportions, δ's	Response Probabilities, ρ's: Nondepressed	Response Probabilities, ρ's: Depressed	Latent Transition Probabilities, τ's: t to t + 1, Class 1	t to t + 1, Class 2	t = 1 to t = 4, Class 1	t = 1 to t = 4, Class 2
Class 1	.805 (.030)	.941 (.008)	.059 (.008)	1.000	.000	1.000	.000
Class 2	.195 (.030)	.372 (.046)	.628 (.046)	.128 (.043)	.872 (.043)	.338	.662

*A parameter value of .000 or 1.000 indicates convergence to a boundary value.

TABLE 6: Estimated Parameter Values (and Standard Errors) for Partially Latent Mover-Stayer Model, G**

Chain, Class	Chain Proportions, π's	Class Proportions, δ's*	Response Probabilities, ρ's: Nondepressed	Response Probabilities, ρ's: Depressed	Latent Transition Probabilities, τ's: t to t + 1, Class 1	t to t + 1, Class 2	t = 1 to t = 4, Class 1	t = 1 to t = 4, Class 2
Chain 1	.985 (.007)							
Class 1		.771 (.050)	.950 (.011)	.050 (.011)	1.000	.000	1.000	.000
Class 2		.229 (.047)	.481 (.075)	.519 (.075)	.137 (.046)	.863 (.046)	.359	.641
Chain 2	.015 (.007)							
Class 1		0.000	1	0	1	0	1	0
Class 2		1.000	0	1	0	1	0	1

*The standard errors of the δ's were obtained from the standard error of $\pi_1\delta_{i|1}$, which is PANMARK output, by considering, π_1, which is near 1, as a constant: $\sigma(\pi_1\delta_{i|1}/\pi_1) = \sigma(\pi_1\delta_{i|1})/\pi_1$.
**Parameter values of 0 and 1 area fixed by definition. A parameter value of .000 of 1.000 indicates convergence to a boundary value.

movers (chain 1) are classified to be nondepressed at t = 1. The off-diagonal transition probabilities (both from t to t + 1 and from t = 1 to t = 4) show that the odds of changing from depressed to non-depressed are about 3:1 as compared to changing from nondepressed to depressed at later points in time.

Because the Mover-Stayer model fits well, it may be advisable to look for more parsimonious models. One such model is the so-called three-parameter model considered by Morgan et al. (1983). It assumes that the probability of changing from time t to time t + 1 does not depend on the category a person belonged to at time t and therefore requires the rows of the mover transition matrix to be equal ($m_{01} = m_{11}$ in the Morgan et al. [1983] terminology). As we mentioned earlier, this model is equivalent to a classical Latent Class model in which the mover initial probabilities may vary freely but response probabilities at all other points in time are constrained to be equal for movers. We thus gain 1 df, and the fit of this model (model B), as well as the difference in fit as compared with model A ($LR_B - LR_A = 2.37$, df = 1), is acceptable.

Because the mover initial probabilities and transition probabilities look rather similar in model B, we finally require these two sets of parameters to be equal. In other words, it is assumed that the movers are in a steady state, or equilibrium at t = 1 already. This is a classical LC model that requires the marginal distributions of the movers to be equal at all four points in time. As the results show, this model (model C) is too restrictive. The fit is not acceptable and there is a significant deterioration in fit as compared with model B: $LR_C - LR_B = 6.25$, df = 1. Comparison of models B and C is in line with an observation often made in panel analysis: The marginal distribution at t = 1 differs from those of the later points in time that tend to be almost equal.

Up to this point, we have considered the Mover-Stayer model and some constrained versions of this model. Instead of imposing constraints, we may proceed in the opposite direction and let the stayer transition probabilities vary freely. The resulting model then is a two-class (or two-chain) Mixed Markov model. In contrast to Mover-Stayer models, this model does not assume perfect stability but a mixture of two (possibly different) Markov processes. This model (model D) not only fits extremely well, but also shows a significant gain in fit over the Mover-Stayer model A ($LR_A - LR_D = 6.96$, df = 2). According to this model, class 1 contains 85.4% of the respondents, and the odds of belonging to category 0 versus category 1 are about 9:1 at t = 1. There is a very high repeat probability for category 0 and a very low one for category 1.

The pattern for class 2 is very different from that of class 1. First, about two-thirds of the members of this group are classified to be

depressive at t = 1. Second, transition probabilities show that there seems to be a considerable change across time in both directions (becoming depressed and becoming nondepressed).

Because the rows of the transition matrices are similar within classes and also similar to the initial probabilities under model D, we should consider a model that assumes (1) that change is independent of the category a person did belong to at a previous point in time and (2) that the probability of being classified depressed is constant across time within classes. This model (model E) thus reduces to a classical two-class Latent Class model in which all response probabilities are constrained to be equal within classes. In other words, no change is assumed.

As it turns out, model E has an acceptable fit. Comparison with model D (from which model E is derived by imposing four constraints) shows that the difference in fit between these two models is insignificant ($LR_E - LR_D = 7.86$, df = 4). In terms of parameters, model E is as parsimonious as model C, but model E obviously gives a better description of the data. As the estimated parameter values show, class 1 is well defined (the vast majority are nondepressive persons). Class 2, on the other hand, is a mixture of nondepressive (40%) and depressive (60%) persons.

As pointed out earlier, all models considered so far assume class membership to be constant through time. The model considered next—the Latent Markov model, called model F in Tables 3 and 5—does not impose this restriction. This model has an excellent fit. In addition, comparison with the no-change model E (which may also be considered as a Latent Markov model without latent change) reveals that the difference ($LR_E - LR_F = 8.08$, df = 2) is significant. We may conclude that some real changes take place across time.

This model says that 80.5% of the respondents are found in class 1 at t = 1, the vast majority of which are nondepressed persons, and that the probability of leaving this class at later points in time is 0. Class 2, on the other hand, mainly contains depressed persons at t = 1. Roughly 13% of these change to class 1 at each of the consecutive time points.

Up to this point we have considered the Mixed Markov model, the Latent Markov model, and some special cases of these two models. We have not yet considered the general LMMC model (3). For the depression data, this general model is hardly of interest because the one-chain two-class Latent Markov model, F, fits the data well. Extension to two latent chains with two classes each would mean an additional six parameters as compared with model F. However, improvement in fit over model F is impossible for such a model on statistical grounds alone (at the 5% level an additional six parameters would require an improvement in the chi-squared value of 12.6, but we have only a deviance of 8.32 left).

The same comment holds for two LMMC models that introduce some constraints on the parameters of the second chain. In one of these cases, the transition matrix of the second chain may be requested to be the identity matrix. This model would say that we assume latent change for the first chain, but no latent change for the second chain. It may be called a latent Mover-Stayer model with measurement error for both movers and stayers. Alternatively, we may assume a model with measurement error for the first chain but without measurement error for the second chain. This would require fixing the response probabilities of the second chain at 1 and 0 (see Table 6). This model would postulate a mixture of one latent chain and one simple chain. Both of these models require estimation of an additional four parameters as compared with model F. On statistical grounds, they share the same fate as the unconstrained two-chain two-class general LMMC model. The only model that might be worth considering is a model in which the constraints of both of the latter two models are combined (see Table 6).

The result (model G in Tables 3 and 6) may be called a partially latent Mover-Stayer model, where the stayers (chain 2) are defined in the same way as in the Blumen et al. (1955) Mover-Stayer model, whereas measurement error and latent change is incorporated into the mover part (chain 1). Technically, this model may be fitted by specifying a $C' = 4$-class Latent Markov model (see Table 2), with additional parameter fixations on the response probabilities and transition probabilities of the second chain. The fit of model G is acceptable. Note that this is an improvement in fit over the standard Mover-Stayer model A ($LR_A - LR_G = 15.50 - 5.16 = 10.34$, df = 2). However, there is no significant improvement in fit over the Latent Markov Model F ($LR_F - LR_G = 8.32 - 5.16 = 3.16$, df = 2). In terms of parsimony, we should prefer model F. Nevertheless, we will take a closer look at the estimated parameter values of model G.

According to this model, 1.5% of the individuals are classified to be perfectly stable stayers, all of them being depressed persons (chain 2). Note that this percentage differs dramatically from the corresponding one under model A. The rest of the persons, 98.5%, are found in chain 1. Because the pattern of estimated parameter values of this chain is very similar to that of model F, we will not go into details.

STABILITY AND CHANGE

If repeated measurements play a role in a research project, substantive interest may center on stability, change, or both. To give some examples: If the focus is on test theoretical issues, we would be interested in stability (and reliability). If some treatment is given between time points, we would be interested primarily in change. Buying behavior, is an example in which the researcher may be interested in both those people who stay with some brand and those who switch to some other brand.

Models considered in this article differ with respect to types of stability and change. As will be shown below in more detail, we may distinguish such quantities as perfect, nonperfect, and true stability and change. It seems worthwhile, therefore, to compare the different models in how they fare with respect to these quantities, which are summarized in Table 7.

Let us first take a closer look at models without an error of measurements interpretation—that is, at Mixed Markov models A through E. In MM models, the total proportion of stability, TOS_{MM}, across T points in time is given by

TABLE 7: Estimated Proportions of Stability and Change in Models A through G

				*Model**				
	A	*B*	*C*	*D*	*E*	*E'*	*F*	*G*
Perfect stability	.525	.554	.553					.015
Stability	.148	.119	.120	.667	.665		.934	.904
True stability						.661	.651	.629
Error						.339	.283	.275
Change	.327	.327	.327	.333	.335		.066	.081
True change							.025	.024
Error							.041	.057
Total error						.339	.324	.332

*Model E' refers to an error-of-measurements interpretation of model E.

$$\mathrm{TOS}_{\mathrm{MM}} = \sum_{s=1}^{S} \pi_s \sum_{i=1}^{R} \delta_{i\,|\,s} (\tau_{j\,|\,is})^{T-1} \qquad (j = i) \tag{4}$$

Total change amounts to TOC = 1.0 – $\mathrm{TOS}_{\mathrm{MM}}$. Note that in principle both $\mathrm{TOS}_{\mathrm{MM}}$ and TOC may be broken down into perfect and nonperfect stability and change, respectively. If the transition matrix of some chain is forced to be the identity matrix (as in models A through C) this chain contains perfectly stable stayers only. Likewise, any transition matrix with parameters fixed at 1 and 0 – except for the identity matrix – would force a chain to contain perfect changers only.

According to model A, we have 52.5% perfectly stable stayers (chain 2) plus .475 ($.678 \times .764^3 + .322 \times .299^3$) = 14.8% nonperfect stayers (chain 1) and 32.7% changers. Note that the rows of the mover transition probabilities from t = 1 to t = 4 are equal (see Table 4). This need not necessarily be the case. It simply says that change across time has reached a steady state at t = 4. Net change from t = 1 to t = 4 is therefore given by .748 – .678 = 7%. The model thus says that there is a slight net change in the positive direction (becoming nondepressed).

For models B and C, proportions of change are equal to those of model A, but perfect stability is estimated to be slightly higher. In analogy with quasi-Latent Class models, models A, B, and C may be called quasi-Mixed Markov models blanking out (i.e., fitting perfectly) the two extreme cells with observed response patterns 0000 and 1111. In fact, models B and C reduce to quasi-Latent Class models. For these models, total proportions of stability and change are therefore equal to the respective observed proportions in the data when all points in time are considered. Finally, net change according to model B amounts to .733 – .659 = 7.4% (becoming nondepressed), whereas net change is 0 in model C by definition. In Markov terminology, the proportions of change in both directions are the same and thus cancel out when net change is considered. This model may therefore be called a no-net-change model.

Models D and E differ from models A, B, and C by not postulating perfect stability. Nevertheless, total proportions of stability and change come close to those of models A, B, and C. Under model D, net change is in the direction of becoming nondepressed for both chains: .936 – .908 = 2.8% (chain 1) and .407 – .342 = 6.5% (chain 2). Note that chain 2 contains persons classified as depressive at t = 1 in the majority. Total net change (9.3% is thus predicted to be slightly higher as compared with models A and B. Model E, again, is a no-net-change model.

We now switch to models incorporating measurement error – that is, to Latent Markov models F and G. In LM models, the total proportion of latent stability, $\mathrm{TOS}_{\mathrm{LM}}$, across T points in time is given by

$$\mathrm{TOS}_{\mathrm{LM}} = \sum_{a=1}^{C} \delta_a (\tau_{b\,|\,a})^{T-1} \qquad (b = a) \tag{5}$$

As a consequence, total latent change is equal to $\mathrm{TOC}_{\mathrm{LM}}$ = 1.0 – $\mathrm{TOS}_{\mathrm{LM}}$. Due to the error of measurements interpretation, both $\mathrm{TOS}_{\mathrm{LM}}$ and $\mathrm{TOC}_{\mathrm{LM}}$ may be partitioned into an error-free (i.e., true) part and an error part by conceiving of the modal response probabilities of each class as reliabilities and the nonmodal ones as error rates. True stability, TRS, is thus given by

$$\mathrm{TRS} = \sum_{a=1}^{C} \delta_a (\tau_{b\,|\,a})^{T-1} (\rho_{i\,|\,a})^{T} \qquad (i = b = a) \tag{6}$$

TABLE 8: Setup of Expected Counts in 2-Class LM Model for 2-Categorical Variable and 2 Waves

			Column	1	2	3	4	5
	Observed Responses		*Latent scores*					
			at t = 1:	*1*		*2*		
Row	*t = 1*	*t = 2*	*at t = 2:*	*1*	*2*	*1*	*2*	
1	1	1		x				
2	1	2			y			
3	2	1				y		
4	2	2					x	
5								N

The error proportion of stability is equal to ERS = TOS_{LM} – TRS. Finally, true change, TRC, is given by

$$TRC = \sum_{a=1}^{C} \sum_{b=1}^{C} \sum_{c=1}^{C} \delta_a \rho_{i|a} \tau_{b|a} \rho_{j|b} \tau_{c|b} \rho_{k|c} \qquad [7]$$

$$(i = a, j = b, k = c, \text{not } a = b = c)$$

and the error proportion of change is equal to ERC = TOC_{LM} – TRC.

Although equations (5) through (7) provide a unique definition of the quantities of interest, the full table of expected counts may allow for a better insight. In general, this table is of size $R^T \times C^T$; that is, of size 16 × 16 for the depression data and the LM model F. We therefore do not reproduce this table but demonstrate the details by referring to a two-class LM model for a two-categorical variable and two waves. The resulting table will be of size 4 × 4 (see Table 8).

Rows 1 through 4 refer to the R^T observed response patterns and columns 1 through 4 contain the C^T combinations of latent classes at the two waves. Rows would sum to total expected counts under the model (column 5). Sums of column sums (row 5) are equal to the respective latent distributions at the two points in time. Columns 1 and 4 contain the total proportion of latent stability. True stability is found only in those cells that have identical row and column patterns (sum of the x's), whereas the remaining cells in columns 1 and 4 are considered as error because at least one error rate (nonmodal ρ) enters into the computation of the respective expected counts. Total latent

change is found in the remaining columns (2 and 3). Again, true change is captured only by those cells that have identical row and column patterns (sum of the y's), and the remaining cells of these columns sum to the error proportion in change.

According to (5), we obtain $TOS_{LM} = .805 \times 1.0^3 + .195 \times .872^3 =$ 93.4% and a total change at 6.6%. True stability is estimated to be $TRS = .805 \times 1.0^3 \times .941^4 + .195 \times .872^3 \times .628^4 = 65.1\%$. We obtain an error proportion in stability of 28.3%. With respect to the partitioning of total change into true change and error, we will not go into details (because computations according to (7) would require several equations – 14, which reduce to 3 in the present case because there is change from class 2 to class 1 only), but simply refer to the results in Table 7 that show true change and the error proportion in change to be 2.5% and 4.1%, respectively. Model F thus says that true stability, true change, and error amount to 65.1%, 2.5%, and 32.4%, respectively. Finally, we want to point out that all change is in the positive direction (becoming nondepressed). The latent proportion of class 1 at t = 4 is given by .805 × 1.0 + .195 × .338 = .871.

As we mentioned earlier, the LM model reduces to a Latent Class model when the transition matrix is fixed to be the identity matrix. As a consequence, the C^T latent classes in Table 8 reduce to C classes, and the quantities that may be considered are true stability and error only. Model E, which has been referred to as a constrained MM model above, may thus also be given an error of measurements interpretation (see Goodman, 1962, 1974). The resulting quantities are listed in column E′ of Table 7.

Finally, the respective quantities for the partially latent Mover-Stayer model G, which allows for perfect stability in addition to model F, are given in the last column of Table 7.

The general conclusions from Table 7 are quite obvious. According to models E, F, and G (all of which allow for an error-of-measurements interpretation) proportions of true stability are 66.1%, 65.1%, and 64.4% (including 1.5% perfectly stables in the last case). These proportions come rather close to proportions of total stability estimated in models A through E (without an error-of-measurements interpretation). The small proportions of true change estimated by models F and G demonstrate the well-known fact that change is overestimated if the problem of measurement error is neglected (Schwartz, 1985) as is done in models A through E. A caveat about such spurious "changes" due to unreliability or response uncertainty may be found in Coleman (1964b).

DISCUSSION AND COMMENTS

Although there has been a considerable interest in modeling the stochastic processes underlying a set of categorical panel data, our review has shown that developments in the field of Markov modeling have been plagued with several problems. In fact, most of these problems were due to inefficient estimation methods. The first aim of this article was to show that efficient estimation methods have been presented in the 1980s for two classes of models—the Mixed Markov model and the Latent Markov model (as well as some special cases of these). The second aim was to provide a unifying framework: Both Mixed Markow models and Latent Markov models are related to the more general Latent Mixed Markov model, which simply extends the one-chain Latent Markov model to an S-chain model. In principle, this model opens a wide avenue for Markov models not considered in the literature so far. The partially latent Mover-Stayer model referred to in the section on definition and fit is but one example.

However, this generality comes at a price. Identifiability is an issue with these more general models. Whereas we know of hardly any problems in this respect with the one-chain Latent Markov model, we have observed such problems for Mixed Markov models with several data sets. Therefore, identification problems may occur with the more general LMMC model.

One way to check the identifiability of a model for a specific data set is to check whether the matrix of second-order derivatives of the log-likelihood with respect to all free, independent parameters (i.e., the information matrix) is of full rank. Even if the information matrix is of full rank, it may be badly conditioned; that is, the ratio of the largest eigenvalue and the lowest eigenvalue of this matrix, the conditioning number, may be large, indicating that the corresponding parameters have large standard errors.

In case of doubt about identification, a safe but tedious strategy is to test whether different sets of starting values yield the same solution. For MM models and LMMC models without turnover between chains, this strategy is strongly advised, not only to check identification, but also because we found two optima of the likelihood for several data sets with two-chain models. This problem does not occur with the Mover-Stayer model and more restricted models.

To demonstrate the implications and features of the Mixed Markov model and the (one-chain) Latent Markov model as well as of some special cases of these, a substantive example frequently used in the Markov-type literature was considered in more detail for six models. The following comments seem to be in order here:

1. Although several authors have at least imagined the extension from the one-chain Markov model to a mixture of two or more of such chains, only Poulsen (1982) presented a general solution. The rich material in Poulsen's dissertation has never been published in a journal, however. Note that Poulsen was also first in presenting an efficient solution of the parameter estimation problem in the Latent Markov model, whereas the articles by Bye and Schechter and Van de Pol and De Leeuw on the same topic appeared in official journals in 1986.

2. Although conceptually interesting, the Mixed Markov model suffers from the fact that it (1) assumes class (chain) membership to be constant across time and (2) does not allow for measurement error. These assumptions are relaxed in the Latent Markov model, which allows us to pinpoint such quantities as true stability, true change, and error. As a consequence, proportions of change will be overestimated by Mixed Markov models as compared with Latent Markov models if data are error-polluted, which is likely to be true in nearly all instances.

3. Some of the models considered end up with a better fit as compared with the Mover-Stayer model than we started off with. The most prominent case here is the Latent Markov model despite the fact that the number of parameters to be estimated is identical for both

models (dichotomous case). This observation fits with one of the arguments of Davies and Crouchley (1986), which led these authors to recommend that the Mover-Stayer model "requiescat in pace." Despite this recommendation, we feel the Mover-Stayer model to be conceptually interesting. In general, it ends up with a dramatic improvement in fit over the simple Markov model by adding R (= number of categories) parameters to the latter one only (LR = 80.08 and df = 12 for the simple Markov model and the depression data). This improvement is simply due to the fact that the extreme cells (with index values 0000 and 1111 here), which in general capture most of the interaction present in a table, are fitted perfectly in the Mover-Stayer model. Nevertheless, this model should be confronted with rival hypotheses that may lead to different conclusions.

4. Several of the models considered in this article fit the depression data well. We do not advocate choosing between these models just on the basis of their fit to the data because it might be capitalization on chance to make a definite choice between them on the basis of this medium-size sample only. Each of these models is a legitimate way to look at the data. One or the other may be better suited in a specific research problem. Attention should be given, however, to the problem of spurious "change."

The extension to the general LMMC model presented in this article is but one of several extensions that might be thought of. We will therefore finally give a brief sketch of some other lines of development that will be presented in detail in another article. In an early section, we mentioned the proposal to attack the problem of population heterogeneity by stratifying the total sample into subsamples according to one or more discrete exogenous variables. Such subsamples may not only be submitted to separate analyses, but also and preferably to a simultaneous analysis. On the one hand, complete heterogeneity of subsamples may be allowed for, whereas, on the other hand, complete homogeneity may be requested. In addition, many (partially homogeneous) models may be thought of that fall in between these two extremes. This development is not restricted to the simple Markov model but, in principle, may be applied to all models considered here and will be much in the spirit of the Clogg and Goodman (1984, 1985, 1986) work on simultaneous Latent Class analysis. Other examples that may be thought of are Mixed Markov models with latent change and higher-order Markov chains. Although higher-order Markov models have a rather bad image in the literature (e.g., Coleman, 1964a; Poulsen, 1982), they may be justified in some cases.

NOTE

1. PANMARK is a user-friendly program for MM and LM models and simultaneous analysis of several subpopulations, written in Turbo Pascal. PANMARK has many options: several standard models and standard restrictions, input of a restrictions file (any set of parameters can be restricted to be equal or fixed), standard deviations, correlations between parameters, first-order derivatives of the log-likelihood with respect to each independent parameter, and an identification test. PANMARK is available on 5.25 diskette for PC, XT, AT under DOS 3.x from the Netherlands Central Bureau of Statistics for dfl.98 (about $47) plus costs (about $13), including a user manual.

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